

Project 1 — Stepping stone 1

EDA and a first model

Due: Wednesday, September 9, 11:59 p.m.
Counts toward: [Project 1](#) (not a lab grade)
Data: [properties_modeling_v4.csv](#)
Column dictionary: [Data card](#)
Starter notebook: [starter_01.ipynb](#)

Open in Colab

[Open in Colab](#)

Upload [properties_modeling_v4.csv](#) to your Colab session (or download from the link above).

Background: where this project sits in the grant framework

California is developing a **public wildfire catastrophe model** under [SB 429](#). The California Department of Insurance (CDI) describes the overall system in the [Public Wildfire Catastrophe Model Fact Sheet](#). Technical expectations for insurers and modelers appear in CDI’s [Wildfire Catastrophe Model Checklist](#) (PRID, updated February 2025).

A full catastrophe model is a **pipeline**, not one regression. At a high level:

Module	Question it answers	Examples in production models
Exposures	What is at risk, where, and under what policy terms?	Geocoded homes, replacement cost, limits, deductibles
Hazard / event generation	How often and how intensely can fire reach a location?	Fuels, weather, fire simulators, burn probability, stochastic event catalogs
Vulnerability	Given fire exposure, how much physical damage?	Damage vs flame length / embers given construction, defensible space
Financial / actuarial	Given damage, what insured loss?	Policy rules, aggregation, loss costs for rate filings

$$\text{Exposures} \rightarrow p(\mathcal{F} \mid x) \rightarrow p(d \mid s, \mathcal{F}) \rightarrow \mathbb{E}[\text{loss} \mid d, \pi]$$

where x is location/value, s is structure characteristics, $\mathcal{F}$ is fire exposure in a specific event, d is damage, and π is policy terms.

Weather and climate enter in the **hazard** module (they drive simulators and event catalogs). They are **not** usually fed directly into the vulnerability model.

This stepping stone is a **small slice** of the **vulnerability** module:

CDI module	In your CSV	In stone 1
Exposures	latitude, longitude, assessed_improved_value, county, city, ...	Explore in EDA; optional features
Hazard (landscape)	fhsz_category, wrc_*	Optional features (static hazard summaries)
Hazard (event)	fire_inside_perimeter, fire_distance_km, fire_perimeter_acres	Optional features (this fire’s footprint)
Vulnerability	structure_*, distances, outbuildings, ...	Primary predictors of serious_damage / damage_category

We are estimating **damage severity** given structures already inspected in or near a wildfire.

Overview

In this stone you will work on a single prepared CSV to understand the outcome, explore features, choose a simple model and loss, train on **train**, and report on **selection**. Do **not** use the **assessment** split yet — that is reserved for stepping stone 2.

$$\mathbb{P}(\text{serious_damage} = 1 \mid \text{fire contact context}, s, x, \mathcal{F}_{\text{event}})$$

serious_damage = 1 if DINS labeled the structure **Major** or **Destroyed**; 0 otherwise.

Recommended splits

Split	Your job
<code>train</code>	Fit models
<code>selection</code>	Compare models
<code>assessment</code>	Assess final selected model (stone 2 only)

Splits are **incident-grouped** (one wildfire does not appear in two splits). Do not reshuffle.

Part A — Exploratory analysis

A1. Baseline damage rates

Using the full table, compute a table of counts and proportions for `damage_category` (all five levels).

A2. Univariate feature scan

For **every candidate feature column** (all columns except identifiers, `split`, and the four outcome columns — see [data card](#)):

- type (numeric vs categorical);
- missingness count and %;
- for numeric: mean, std, min, max (on `train`);
- for categorical: number of levels and top 5 levels (on `train`).

Present as a compact table printed out in your notebook.

A3. Choose three features for deep dive

Pick **three** features you might use in a model (mix of s , x , H , or $\mathcal{F}$ if possible). For each:

1. **Original source** — one paragraph with a link to the primary data documentation (start from the [data card](#) source table; follow through to DINS field definitions, FRAP FHSZ, WRC, or CAL FIRE perimeters as appropriate).
2. **Missingness** — who is missing? Is missingness associated with `serious_damage` or `damage_category`?
3. **Bivariate relationship** — plots or tables vs `damage_category` and vs `serious_damage` (e.g. boxplots, grouped bar charts as appropriate).

Part B — First model (LM workflow)

Follow this modeling checklist from lecture (problem → data → model → loss → fit → evaluate):

B1. Problem and label

Restate the problem your model will solve, including the unit, class label, and feature summary.

B2. Model choice

Choose **one simple model** you can train, for example:

- logistic regression;
- a tree-based model;
- a method from DATA 322;
- another method you can justify.

State **input features** (subset of columns), how you encode categoricals, and how you handle missing values (rules fit on **train** only).

B3. Loss and metric

- **Loss for training** (e.g. BCEWithLogitsLoss for binary `serious_damage`).
- **Choose a metric for model comparison** on **selection** (e.g. Brier score, ROC-AUC, accuracy).

B4. Train and evaluate

1. Fit on **train** only.
2. Report training loss curve or final training metric.
3. Evaluate on **selection** — report your primary metric.

What to submit

One notebook export containing your answers to parts A and B.